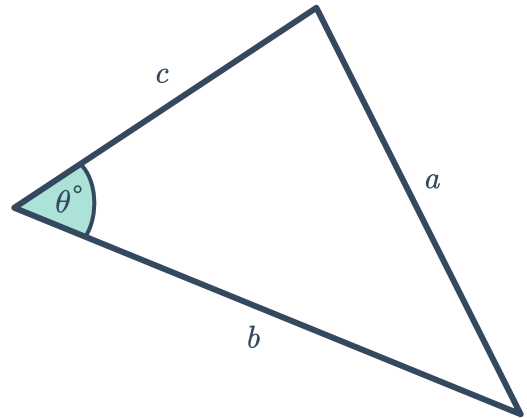


Cosine rule

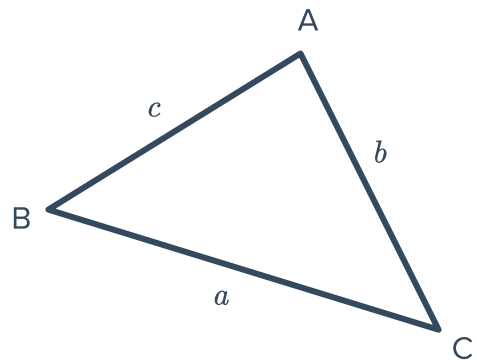


- 1 Write an expression for $\cos \theta$ using the cosine rule for the following triangle:



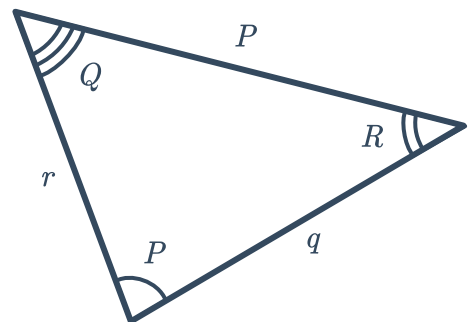
- 2 Consider the triangle given below:

- a Write an expression for $\cos A$ using the cosine rule.
- b Write an expression for a^2 using the cosine rule.
- c Write an expression for b using the cosine rule.

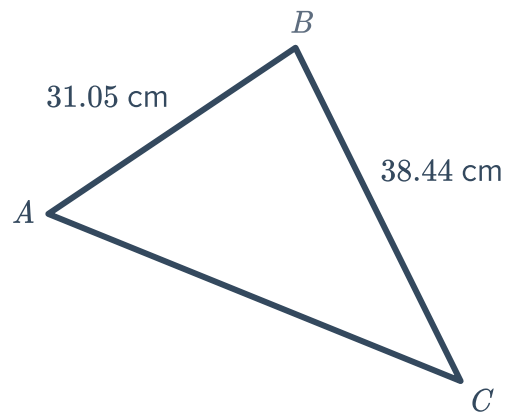


- 3 Consider the triangle given below:

- a Write an expression for $\cos Q$ using the cosine rule.
- b Write an expression for r^2 using the cosine rule.



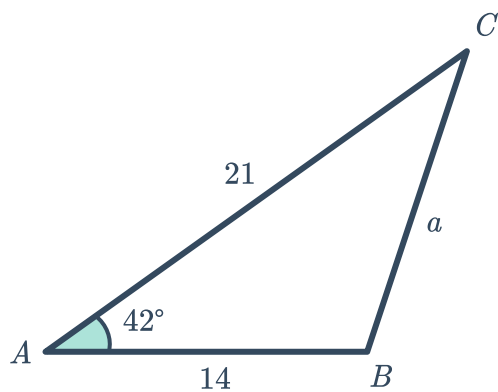
- 4 To use the cosine rule to find the length of AC , which angle would need to be given?



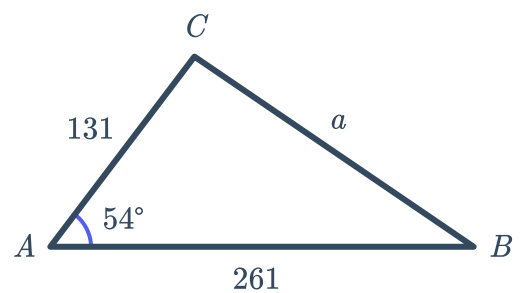
Unknown sides

- 5 Find the length of the missing side in each of the following triangles using the cosine rule. Round your answers to two decimal places.

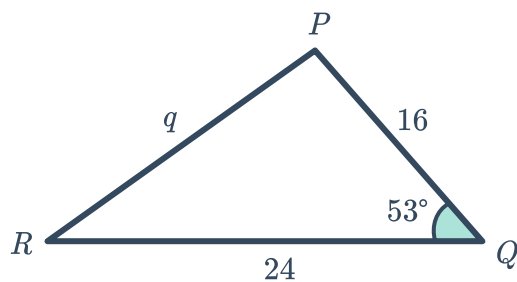
a



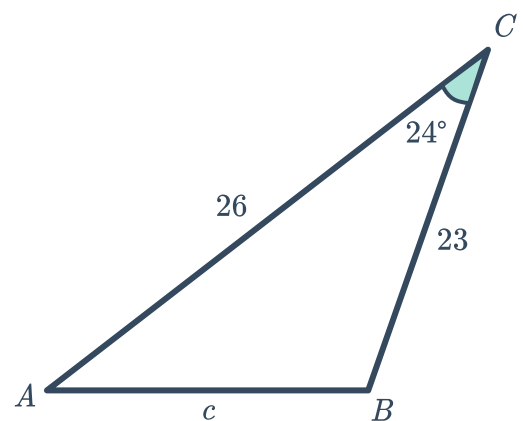
b



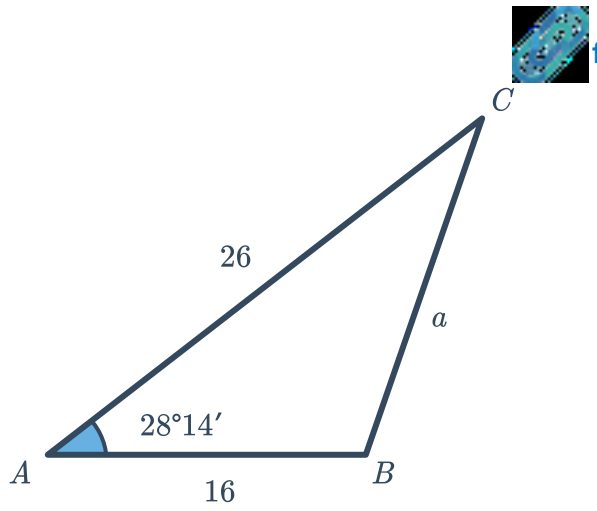
c



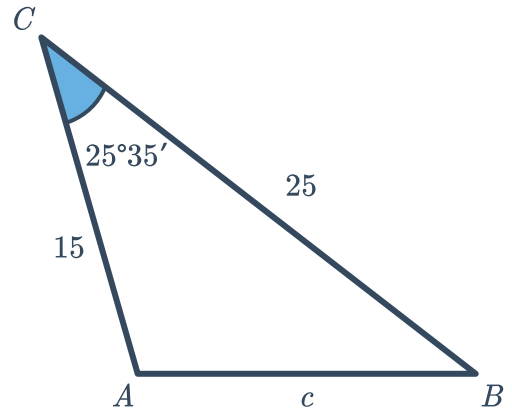
d



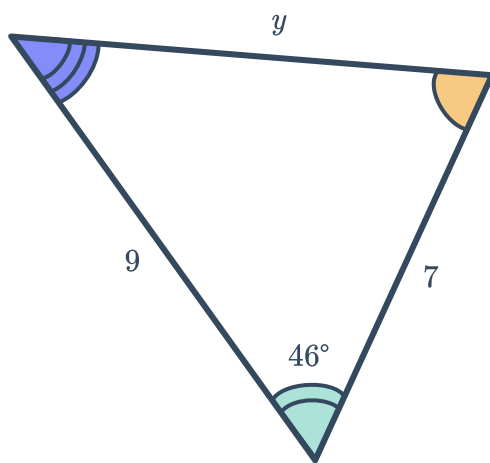
e



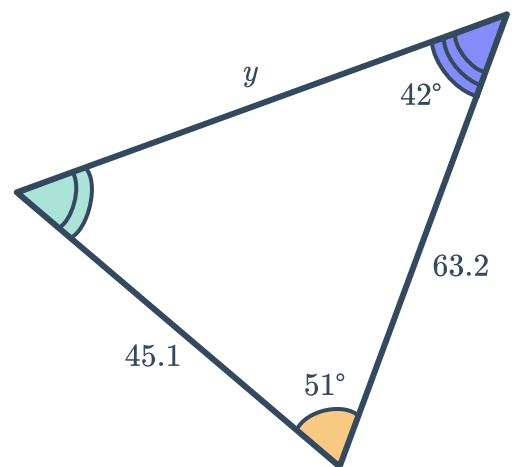
f



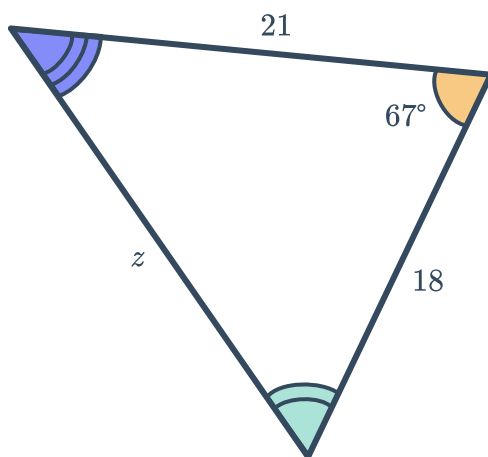
g



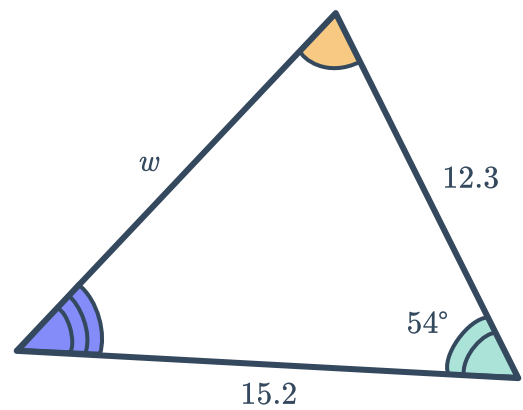
h



i

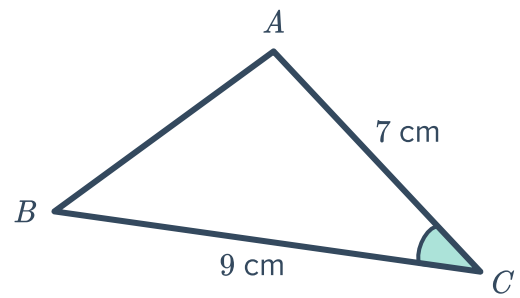


j



6 In $\triangle ABC$, $\cos C = \frac{8}{9}$:

Find the exact length of side AB in centimetres.

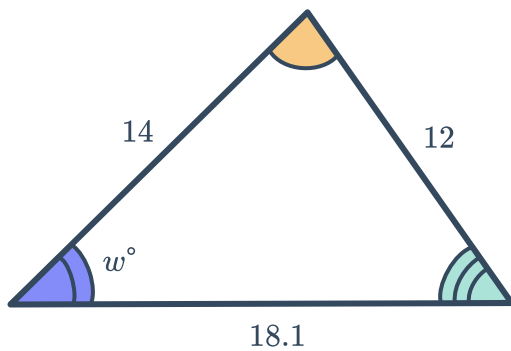


- 7 In $\triangle QUV$, $q = 5$, $u = 6$ and $\cos V = \frac{3}{5}$. Find the value of v .

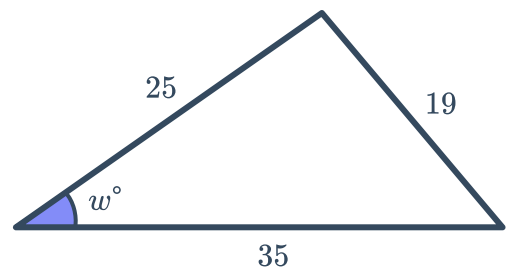
Unknown angles

- 8 For each of the following triangles, find the value of the pronumeral to the nearest degree:

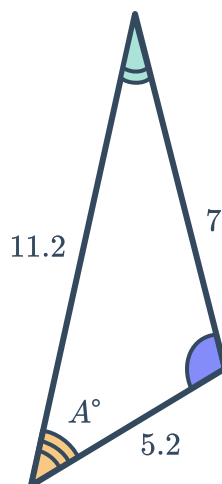
a



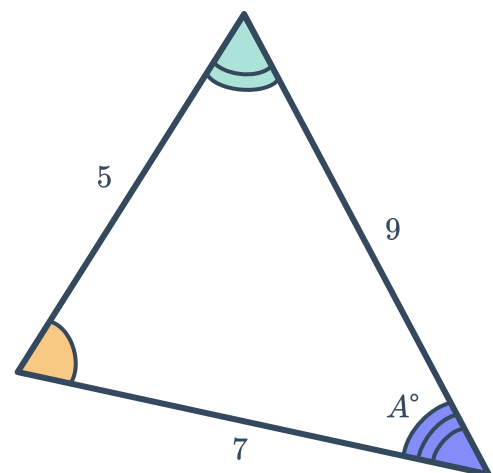
b



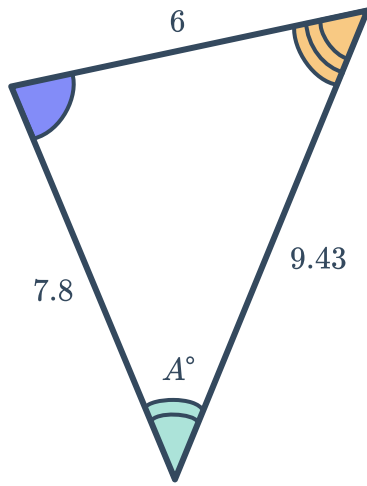
c



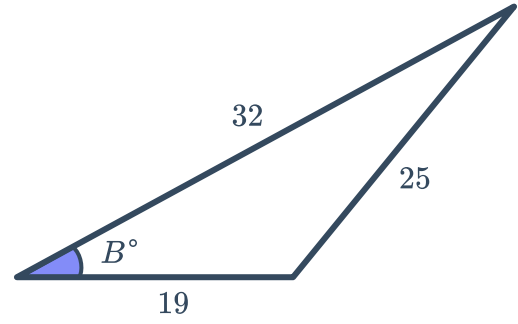
d



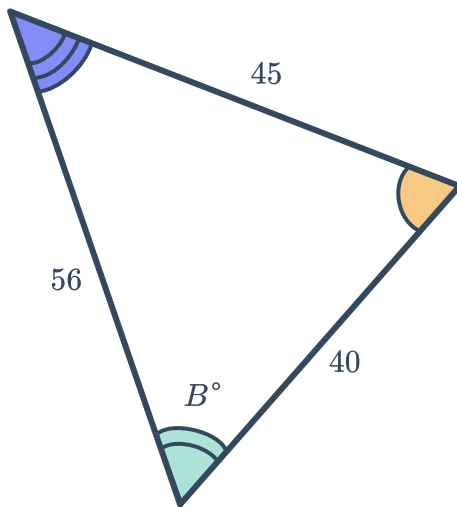
e



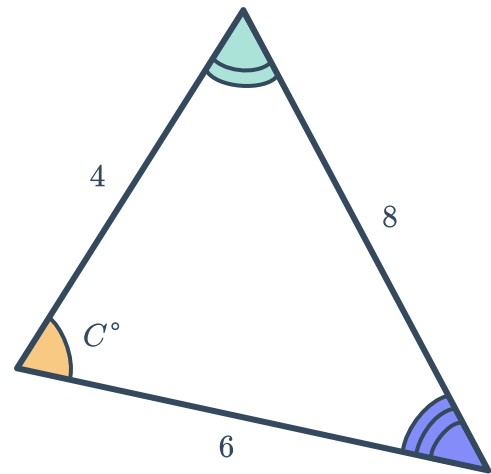
f



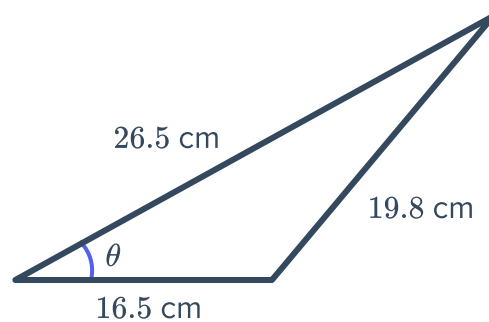
g



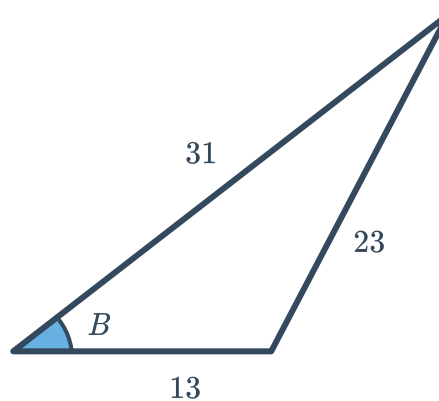
h



- 9 Find the value of θ in the following triangle. Round your answer to the nearest hundredth of a degree.



- 10 Find the value of B in the following triangle. Round your answer to the nearest second.



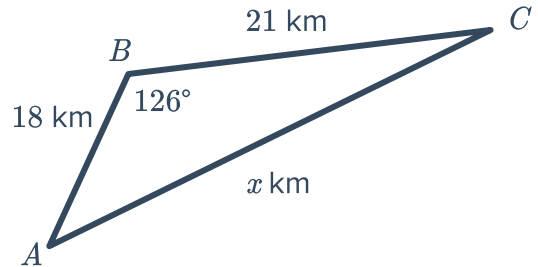
- 11 In $\triangle QUV$, $v = 8$, $u = 9$ and $q = 15$. Solve for $\cos Q$.
- 12 The sides of a triangle are in the ratio 4 : 5 : 8. Find θ , the smallest angle in the triangle, to the nearest degree.
- 13 A triangle has sides of length 11 cm, 18 cm and 8 cm. Find x , the largest angle in the triangle, to the nearest degree.



Applications

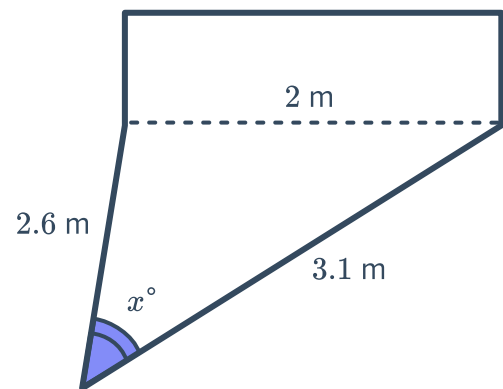
- 14** Mae went for a bike ride on Sunday morning from Point A to Point B , which was 18 km long. She then took a 126° turn and rode from Point B to Point C , which was 21 km long.

Find x , the distance in kilometres from her starting point to Point C to two decimal places.



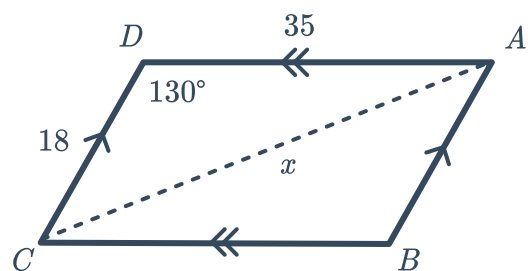
- 15** A goal has posts that are 2 m apart. Buzz shoots for the goal when he is 2.6 m from one post and 3.1 m from the other post.

Find the size of the angle, x , in which he can score a goal. Round your answer to the nearest degree.



- 16** Find the length of the diagonal, x , in parallelogram $ABCD$.

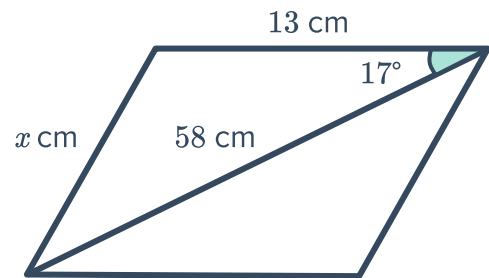
Round your answer to two decimal places.



- 17 Consider the parallelogram in the given diagram that has a side of length 13 cm and a diagonal of length 58 cm:



Find the value of x . Round your answer to one decimal place.



- 18 In a sailing boat race, teams must start at buoy A and navigate around buoys B and C before returning to buoy A to cross the line. The first leg of the race is 100.6 km long, the second leg of the race is 190.3 km long, and the angle between these legs is 143° .

- a Find x , the distance of the third leg of the race, correct to one decimal place.
- b Hence, find the total length of the race, correct to one decimal place.

- 19 A bridge connects two towns on either side of a gorge, where one side of the gorge is inclined at 59° and the other side is inclined at 70° . The length of the steeper incline is 59.1 m.

Find x , the length of the bridge. Round your answer correct to one decimal place.

